Dhiraj Kumar Yadav

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Professional Summary

Backend-focused Software Engineer with 2+ years leading engineering teams and architecting production REST APIs sustaining 100,000+ requests/day at 99% uptime. Certified Claude Architect and AWS Developer with a track record of driving production reliability, optimizing QA workflows (cut manual testing effort by 80%, raised coverage 64% → 95%), and building LLM-integrated products from scratch – both at TCS and through independent full-stack projects (React, FastAPI, LLM APIs).

Certifications

- **Claude Certified Architect – Professional**, Anthropic — proctored, closed-book certification validating the design and leadership of enterprise-scale Claude deployments: agentic architecture, Claude Agent SDK, Claude Code, Model Context Protocol (MCP), multi-agent orchestration, and production reliability/governance at scale.
- **AWS Certified Developer – Associate**, Amazon Web Services — validates skills in developing, deploying, and debugging cloud-native applications using AWS services.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, Java, SQL
- **Backend:** FastAPI, Flask, Django, Node.js, Express.js, REST APIs, Microservices
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Cloud & DevOps:** AWS (EC2, S3, IAM, Lambda, API Gateway, DynamoDB, CloudWatch), Docker, Git, GitHub, CI/CD
- **Databases:** MongoDB, MySQL, SQLite
- **Software Engineering:** Data Structures & Algorithms, OOP, System Design, Distributed Systems, Design Patterns
- **Testing:** PyTest, Postman, API Testing, Debugging, Production Support
- **AI & Agentic Systems:** Claude Agent SDK, Claude Code, Model Context Protocol (MCP), Multi-Agent Orchestration, AI System Design, Google Gemini API, Prompt Engineering, LLM Integration, Structured Output

Professional Experience

Tata Consultancy Services (TCS) Aug 2024 – Present
Software Engineer New Delhi, India

- Architected, built, and maintain 10+ production REST APIs in Python (FastAPI, Flask, Django) powering healthcare workflows, sustaining 100,000+ requests/day at 99% uptime with an average response time of 155ms.
- Serve dual roles: backend developer on the primary healthcare API team, and Tech Lead for a separate 7-member test automation team, planning feature delivery, running code reviews, and mentoring developers as the primary technical point of contact for onsite stakeholders.
- Own backend features from design through production, working directly with product owners and QA to translate requirements into shipped, tested functionality without hand-off gaps.
- Designed and shipped internal test automation tooling that cut manual testing effort by 80% while raising test scenario coverage from 64% to 95%.
- Mentor and onboard new engineers onto the team, cutting ramp-up time to full productivity from 4 weeks to 1.5 weeks.
- Am the first person called for critical production incidents, resolving 3–4 per quarter through log analysis and root-cause debugging while minimizing customer-facing impact.
- Improve application security by implementing authentication and authorization mechanisms, and work with security teams to fix vulnerabilities and meet compliance requirements.
- Built monitoring and operational tooling for distributed backend systems, giving the team earlier visibility into failures before they reached customers.
- Improve code quality across the team through structured production code reviews, making the codebase easier to maintain long-term.

Projects

Inbox CFO *React.js, FastAPI, SQLite, Google Gemini API, Gmail API*

- Designed and built an AI finance agent in a single weekend that reads a user's Gmail transaction history and turns it into categorized spending insights and recommendations, with zero manual data entry.
- Built the FastAPI backend end-to-end: secure email ingestion, expense categorization logic, and a recommendation engine powered by Google's Gemini LLM.
- Implemented OAuth-based Gmail integration so the system reads financial data without ever storing raw email content.

Developer Portfolio (Live) – dhiraj-yadav.in *React.js, TypeScript, Node.js, Express.js, MongoDB, Gemini API*

- Built and maintain a fully dynamic full-stack personal site (initial build completed over a weekend) with an admin dashboard, so all projects and blog content update live from the database – no redeployment ever required.
- Shipped a JWT-secured admin layer for dynamic content management, separating public and authenticated views cleanly.
- Integrated Gemini API to power an in-site AI assistant that answers visitor questions about my experience and projects directly.

Real-Time Chat Application *React.js, Node.js, Express.js, MongoDB, Socket.IO, JWT*

- Built a real-time messaging platform on Socket.IO supporting live delivery, presence, and typing indicators.
- Implemented JWT-based auth and persistent chat history backed by MongoDB for reliable message retrieval across sessions.

Leave Management System *React.js, Node.js, Express.js, MongoDB*

- Built a role-based leave management platform spanning employee, manager, and admin workflows end-to-end in 3 weeks; now used by 800+ people.
- Designed the approval pipeline and attendance-tracking data model, with a dashboard surfacing team-level leave analytics.

Education

Guru Gobind Singh Indraprastha University 2021 – 2024
Bachelor of Computer Applications (BCA) – CGPA: 8.45/10 New Delhi, India